

DATA VISUALIZATION IN PYTHON

Seminar Quiz — Questions & Answers

1. Multiple Choice Questions (MCQs)

1. Which Python library is mainly used for plotting graphs?

- A. NumPy
- B. Matplotlib
- C. Pandas
- D. Seaborn

Answer: B. Matplotlib

2. Which plot shows the relationship between two numerical variables?

- A. Pie chart
- B. Scatter plot
- C. Histogram
- D. Box plot

Answer: B. Scatter plot

3. Which plot is commonly used to visualize a correlation matrix?

- A. Heatmap
- B. Rug plot
- C. Bar plot
- D. Line plot

Answer: A. Heatmap

4. Which plot displays data in three dimensions?

- A. 2D plot
- B. 3D plot
- C. Pie plot
- D. Count plot

Answer: B. 3D plot

5. Which Seaborn plot can show the distribution of a numerical variable?

- A. Histogram
- B. Pairplot
- C. Lmplot
- D. Jointplot

Answer: A. Histogram

6. Which plot helps identify outliers using quartiles?

- A. Box plot
- B. Line plot
- C. Heatmap
- D. Contour plot

Answer: A. Box plot

7. Which library is designed for statistical data visualization?

- A. Seaborn
- B. Requests
- C. Flask
- D. Tkinter

Answer: A. Seaborn

8. Which route represents a shortest path over a spherical Earth?

- A. Straight line
- B. Great-circle route
- C. Grid line
- D. Contour line

Answer: B. Great-circle route

2. Odd One Out

1. Matplotlib, Seaborn, Cartopy, NumPy

Answer: NumPy — the others are commonly used for plotting or geographic visualization.

2. Histogram, KDE, Rug plot, Great-circle route

Answer: Great-circle route — it represents a geographic path; the others can show distributions.

3. Box plot, Violin plot, Strip plot, Heatmap

Answer: Heatmap — the others commonly show distributions across groups or observations.

4. plot3D, scatter3D, bar3d, pairplot

Answer: pairplot — the others are 3D plotting functions.

5. Latitude, Longitude, Map projection, Histogram

Answer: Histogram — the others relate to geographic mapping.

6. Elevation, Azimuth, Camera view, KDE

Answer: KDE — the others relate to setting or describing a 3D viewing angle.

3. Match the Following

Column A

- 1. Matplotlib
- 2. Seaborn
- 3. GeoPandas
- 4. Heatmap
- 5. Pairplot
- 6. Basemap

Column B

- A. Geographic data
- B. Statistical visualization
- C. Multiple pairwise relationships
- D. Mapping toolkit
- E. General plotting
- F. Correlation matrix visualization

Answers: 1–E, 2–B, 3–A, 4–F, 5–C, 6–D

4. True or False

1. A histogram is used to show the distribution of numerical data.

Answer: True

2. A heatmap can visualize correlation values.

Answer: True

3. A 3D scatter plot uses only two axes.

Answer: False — it uses three spatial axes.

4. A box plot can help identify outliers.

Answer: True

5. Seaborn is built on top of Matplotlib.

Answer: True

6. A great-circle route represents a shortest path on a sphere.

Answer: True

7. A KDE plot displays only raw individual data points.

Answer: False — KDE shows a smoothed estimate of a distribution.

8. Every map projection preserves all geographic properties perfectly.

Answer: False — projections involve distortions.

9. A pairplot can show relationships between multiple numerical variables.

Answer: True

10. The viewing angle of a 3D plot cannot be changed.

Answer: False — it can be adjusted.

5. Quick-Fire — One-Word Answers

1. Library used for statistical visualization?

Answer: Seaborn

2. Plot used for correlation visualization?

Answer: Heatmap

3. Plot used to show quartiles?

Answer: Boxplot

4. Coordinate for north–south position?

Answer: Latitude

5. Coordinate for east–west position?

Answer: Longitude

6. Plot showing a 3D point cloud?

Answer: Scatter

7. Technique for estimating a smooth distribution?

Answer: KDE

8. Representation of Earth's surface on a map?

Answer: Projection

9. Plot showing several variable-pairs?

Answer: Pairplot

10. Seaborn plot function used for regression relationships?

Answer: Lmplot